

ELI — Architecture

ELI v2.4.68

September 2026

Contents

Blueprint — ELI Architecture	1
0. Design principles	2
1. Repository map	2
2. Entry points and launch	3
3. The request lifecycle	4
4. Routing layer (eli/execution)	4
5-6. Agent bus and agents (eli/cognition/agent_bus.py)	5
7. The orchestrator (eli/cognition/orchestrator.py)	5
8. Inference layer (eli/cognition)	6
9. Grounding and anti-confabulation (eli/runtime, eli/core/grounding.py)	6
10. Execution layer (eli/execution/executor_enhanced.py)	6
11. Memory subsystem (eli/memory)	7
12. Perception (eli/perception)	7
13. Persona (eli/cognition/persona*)	8
14. Daemons and background loops	8
15. Learning and self-training (eli/learning)	8
16. EliWorld (eli/world, the EliWorld tab, kernel/world_model.py)	8
17. Core infrastructure (eli/core)	8
18. GUI (eli/gui)	9
19. On-disk layout (artifacts/, runtime)	9
20. Known seams and weak points	9
21. Where to make a change	10
Plugins, agent specifications and training surfaces	10

Updated for v2.4.68 (September 2026). Every CHAT turn runs the gradient orchestrator at a depth scaled to the reasoning mode, with the specialist bus composed inside it after shared retrieval (eli/memory/retrieval.py). Canonical S01-S12 tracing is in eli/kernel/pipeline_trace.py; stage 12 learning runs through learning_coordinator.py.

Blueprint — ELI Architecture

The structural map of ELI v2. File paths, counts and names below were checked against the source tree on 2026-09-25; where something is only true at runtime it is marked (*runtime*).

ELI is a **local-first, offline-by-default, model-agnostic** cognitive runtime and assistant GUI, and also a self-hosted web app (api/server.py, described in ELI_USER_MANUAL.md). No cloud on the inference path, no hardcoded model. **193,541 lines across 434 Python files in eli/**, plus api/server.py (2,292 lines). 227 capabilities in capability_manifest.json, 186 of them routable and 205 in the executor's supported list.

0. Design principles

1. **Local-first, offline-by-default.** Outbound network is refused at the socket boundary (`eli/core/netguard.py`) unless the owner enables it; features that need the network opt in.
 2. **Model-agnostic.** No model name or size is hardcoded on the inference path. ELI loads whatever GGUF is configured and identifies its family from file metadata.
 3. **Grounded.** A dedicated subsystem (§9) exists to stop a local model stating things it cannot back up.
 4. **Persona is not scripted away.** ELI has a distinct voice; functional bugs are fixed without flattening it.
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1. Repository map

Files and lines per package under eli/, measured 2026-09-25.

Package	Lines	Files	Role
eli/runtime	35,375	95	Grounding spine, evidence, response and introspection surfaces, daemons, self-improvement
eli/gui	27,447	27	PySide6 desktop app, panels, tabs, startup and first-boot wizard, animated face
eli/execution	26,293	13	Router and executor: intent → action → side effects
eli/cognition	20,824	47	Agent bus, 12-stage orchestrator, inference, persona, reasoning modes, tone and emotion
eli/kernel	17,545	8	CognitiveEngine, pipeline tracing, scheduler, self-upgrade, world model
eli/core	11,222	30	netguard, paths, settings, hardware profile, model download, full control
eli/perception	10,220	23	Vision, STT, TTS, wake word, voice tone, OS control, gaze
eli/memory	8,770	11	SQLite, FTS5, FAISS, knowledge graph, storage policy

Package	Lines	Files	Role
eli/tools	7,573	29	Image engine, news, capability registry, document tools
eli/plugins	5,911	34	Plugin manager, marketplace client, bundled plugins
eli/learning	4,251	14	LoRA self-training pipeline
eli/planning	4,000	19	Proactive daemon, habit scheduler, job, proposal and attention queues
eli/integrations	3,339	18	Ollama client, MCP client and server, cross-platform media, MPRIS
eli/coding	2,104	12	CodeAgent: plan → search → verify → repair
eli/setup	1,935	9	Installer, hardware policy, first-run wizard
eli/utils	1,838	5	Logging, platform compatibility
eli/world	1,793	26	World event bus, local world bridge, avatar and ontology
eli/system	1,190	3	Portable app control, process guard
eli/contracts	787	3	Typed runtime-status contracts
eli/onboarding	709	2	Onboarding interview
eli/cli, top level	236	4	Headless REPL, package entry
total	193,503	434	

The largest files: eli/kernel/engine.py (16,014 lines), eli/execution/executor_enhanced.py (15,845), eli/gui/eli_pro_audio_gui_v2_0.py (13,107), eli/execution/router_enhanced.py (8,271), eli/gui/labs_tab.py (5,744), eli/memory/memory.py (5,677), eli/cognition/agent_bus.py (3,517), eli/runtime/deterministic_grounding_ga (3,367), eli/cognition/gguf_inference.py (3,234).

2. Entry points and launch

Command	Target	Notes
eli, eli-v2.0, eli-cli	eli.gui.app:main	GUI (default)
eli-mcp	eli.integrations.mcp.server:main	ELI as an MCP server
eli-jobs	eli.planning.jobqueue_cli:main	durable job queue CLI

Command	Target	Notes
<code>python -m eli</code>	<code>eli/__main__.py</code>	GUI, or headless with flags
<code>python -m eli --headless</code>	<code>eli.cli.headless</code>	terminal REPL, no Qt
<code>./eli.sh</code>	<code>python -m eli</code>	venv launcher
<code>bash install.sh</code>		venv, dependencies, clean config seed, verification

Boot path (GUI): `eli/gui/app.py` → first-boot wizard when no model is configured (`eli/gui/panels/startup.py`)
→ `eli_pro_audio_gui_v2_0.py:main()` builds `EliMainWindow` → constructs the `CognitiveEngine` singleton
→ loads the GGUF named in `config/settings.json` → starts the daemons.

3. The request lifecycle

Every request goes through `CognitiveEngine.process()` (`eli/kernel/engine.py`).

```

user input
|
▼
[Router] eli/execution/router_enhanced.py :: route()
|   priority pipeline → {action, args, confidence, meta.matched_by}
|   ▼
CognitiveEngine.process(action, args, ...)
|   ├── deterministic fast path (OS, media, status, job actions)
|   |   → execute_action() → returned verbatim, no model
|   ├── CHAT (any reasoning mode, Quick to Expert)
|   |   → AgentOrchestrator.run() at mode depth
|   |       S05 planner → shared retrieval → dispatch_specialists()
|   |       → context assembly → broker.infer() → output governor
|   ├── non-CHAT action
|   |   → AgentOrchestrator → bus + ReAct tool loop
|   └── orchestrator returns None or raises → AgentBus.dispatch() fallback

```

The canonical stage names (`pipeline_trace.STAGE_NAMES`) are S01 PERCEIVE_INGEST, S02 INPUT_GUARDS, S03 ROUTER, S04 GROUNDING_GATE, S05 PLANNER, S06 AGENT_BUS, S07 CONTEXT_ASSEMBLY, S08 INFERENCE_BROKER, S09 REASONING_SYNTHESIS, S10 OUTPUT_GOVERNOR, S11 RESPONSE_DELIVERY, S12 LEARNING_STATE_UPDATE. Enable tracing with `ELI_PIPELINE_TRACE=1` or `scripts/eli_startup.sh --trace`.

4. Routing layer (eli/execution)

- **router_enhanced.py**: `route(text) -> {action, args, confidence, meta}`. A regex-first priority pipeline of ordered prepasses (realtime web lookup, media, file and PDF, memory, control), then a core router, with an LLM intent fallback (`cognition/llm_intent.py`).
- **execution_planner.py**: typed `ExecutionPlan` and `RouteDecision` artifacts.
- **route_contracts.py**, **operator_policy.py**, **shell_gate.py**: what may be routed and executed.
- **portable_intent_contract.py**, **media_runtime.py**, **operator_actions.py**: media and OS intent wiring. Plugin actions are handled directly in `executor_enhanced.py`.

Routes that were once bugs are pinned by tests and by tools/eval/cases.yaml: media controls need whole-word, command-shaped input; a bare “do a search” needs a subject; meta-questions stay in CHAT; realtime facts go to WEB_SEARCH; code words such as “repo” match whole words only, so “morning report” is not a codebase audit.

5-6. Agent bus and agents (eli/cognition/agent_bus.py)

AgentBus.dispatch(user_input, intent, ...) -> DispatchResult selects an agent set (_select_agents_for_intent, or a broad fan-out), runs it over a dependency DAG (flat parallel if the DAG fails) and aggregates. DispatchResult carries agent_results, action_result, memory_context, aggregated_confidence, grounding_confidence (agent evidence only), agents_used, confidence_label and orchestrator_plan.

15 registered agents (_ALL_AGENTS):

Agent	Role
BusMemoryAgent	recall from SQLite, FTS and FAISS into context
SystemAgent	runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)
HabitAgent	learned habits
SelfImprovementAgent	recent failures, improvement proposals, corrections
ProactiveAgent	proactive patterns and signals
FrontierAgent	frontier reasoning hooks
PluginAgent	dispatches to registered plugins
CapabilityAgent	what ELI can do (capability manifest)
VoiceAgent	STT and TTS engine state
OrchestratorAgent	plans and coordinates grounded synthesis
FileCodeAgent	searches the codebase for code-grounded answers
ReflectionAgent	session reflection
IntrospectionBusAgent	live runtime and self introspection
KnowledgeGraphAgent	entities and relations from the knowledge graph
CriticAgent	second-tier verifier; runs after its retriever dependencies

SpecAgent runs user-defined agent specifications against the local model. The coding pipeline, CodeAgent (eli/coding/agent.py, plan → search → verify → repair), is separate from the bus and is invoked for CODE_SOLVE and GENERATE_SCRIPT.

7. The orchestrator (eli/cognition/orchestrator.py)

Runs for all CHAT modes at gradient depth, and for non-CHAT actions. Shared retrieval lives in eli/memory/retrieval.py (retrieve_for_turn()), called by both BusMemoryAgent and the orchestrator so a turn is not searched twice:

- **Turn cache:** 8 seconds per process. Its key includes every retrieval limit, so a deeper pass with larger limits is never served the shallower result.
- **Time window first:** when the question names a period (“last week”, “the past two weeks”), query_planner.parse_window produces a window. Memories inside it are fetched by date (Memory.memories_between) before any ranking, then ranked by topic. The search reports what the window did (memory_diag).
- **Sequential by design:** the llama.cpp embedder is not thread-safe.

- **Heuristic rerank** (cognition/reranker.py): lexical overlap $\times$ recency $\times$ importance. A neural cross-encoder is not in use.
- **FAISS tombstones**: `vector_store.mark_memory_deleted()` marks stale vectors without a rebuild; `compact_tombstones()` reclaims space.

Mode scaling (reasoning_modes.py): Quick uses the fast planner and a lean specialist profile; Normal, Advanced, Research and Expert use progressively deeper planner budgets and wider `mode_chat_agent_profile()` fan-out. Quick still runs the orchestrator.

8. Inference layer (eli/cognition)

- **inference_broker.py**: the canonical inference path, `broker.infer()`.
 - **gguf_inference.py**: the llama-cpp wrapper: live runtime parameters, `chat_completion()`, runtime snapshot publishing.
 - **model_identity.py, model_output_tokens.py**: model family and chat template are read from GGUF metadata, not from the file name; special tokens are stripped per family.
 - **reasoning_modes.py**: quick, chain_of_thought, self_consistency, tree_of_thoughts, constitutional_ai (private strategies; the raw reasoning is not shown).
 - **context_synthesiser.py, user_info_builder.py**: prompt and context assembly.
 - **context_budget.py**: how much of the window memory may use. Output reserve is an eighth of the window; on a recall question the memory floor is a third of it.
 - **llm_intent.py**: LLM fallback for ambiguous routing. Decoding is constrained by a GBNF grammar built from the live action catalogue, so the model can only name a real capability and emit well-formed JSON. Backends without grammar support fall back to free text.
 - Budgeting lives in `engine.py::_build_enhanced_system` and `core/dynamic_runtime_budget.py`.
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9. Grounding and anti-confabulation (eli/runtime, eli/core/grounding.py)

- **deterministic_grounding_gate.py**: decides when an answer must be backed by evidence and renders self-reports and audits from live measurements. `render_action` is one explicit ordered pipeline of layers (`_stack`), not captured copies.
- **grounding_escalation.py**: a checkable factual turn with low grounding escalates (external fact $\rightarrow$ web agent, self or project fact $\rightarrow$ broad local fan-out) and hedges honestly if nothing grounds it. Env: `ELI_GROUNDING_ESCALATION`.
- **evidence_ledger.py, evidence_arbitration.py**: what counts as evidence, where it is recorded (the ledger also records every executed action), and conflict resolution.
- **persistence_gate.py**: keeps internal report dumps out of memory.
- **response_contracts.py, response_policy.py, final_response_provider.py, user_visible_response_surface.py**: shape and clean the user-visible answer.
- **truth_report.py, eli_identity_audit.py, control_contracts.py**: runtime truth, identity grounding, control-action contracts.
- **cognition/output_governor.py, response_governance.py, response_sanitizer.py, persona_hygiene.py**: the final governance pass, including removal of false diagnoses when retrieval demonstrably ran.

The known weak seam is internal state leaking into spoken output on the plain CHAT path (identity confabulation, telemetry dumps). It is defended, not closed.

10. Execution layer (eli/execution/executor_enhanced.py)

- `execute(action, args) -> dict`. The executor's supported list holds **205 actions**; the manifest declares **227 capabilities**, of which **186 are routable** and 205 are in the supported list.

- **Fast path** (engine.py): deterministic OS, media, status and job actions (VOLUME, MEDIA_CONTROL, NEXT_MEDIA, OPEN_APP, DATE, SHELL_EXEC, ANALYZE_IMAGE, CHECK_JOB, BACKGROUND_JOBS, ...) return the executor result verbatim.
- Action families: media and OS control, files and documents (SUMMARIZE_FILE, ANALYZE_PDF[_FOLDER] including per-file mode and saving into memory, CREATE_DOCUMENT), web (WEB_SEARCH), news, weather, memory (MEMORY_STORE, MEMORY_RECALL), code (CODE_SOLVE, GENERATE_SCRIPT), background jobs, self and runtime reports (RUNTIME_STATUS, SELF_REPORT, RUNTIME_AUDIT, EXPLAIN_*), vision, image generation, scheduling.
- **Background jobs** (runtime/background_tasks.py): submit/get/list. Heavy PDF-folder analysis backgrounds itself; CHECK_JOB returns the real result.
- **YouTube playback** lives in eli/integrations/media/youtube_playback.py; the executor delegates to it.

Plugins (eli/plugins)

Bundled: calendar, document_reader, media, notes, pomodoro, system_stats, tts, weather, web, web_automation. State is in config/plugins_state.json (gitignored). Custom agents and plugins are trust-gated (\$ plugin table below).

11. Memory subsystem (eli/memory)

- **memory.py** (Memory, get_memory()): long-term store over SQLite and FTS5, plus failure, observation and habit logging.
- **policy.py**: the storage policy. Every row carries an origin (user_said, eli_said, telemetry, news, tool); repeats are merged (seen_count); a forgetting curve scales weight by importance, origin and reinforcement; faded derived rows move to memories_archive. A daily run_upkeep() does the merging, decay, archiving, identity merge and learning-table tidy. Full detail in memory.md.
- **vector_store.py**: FAISS index at artifacts/vectors/index.faiss; embedder models/embeddings/nomic-embed-text-v1.5.Q4_K_M.gguf (*runtime*).
- **Knowledge graph**: kg_entities and kg_relations (FTS5-backed) in the user database.
- **Working memory** (cognition/working_memory.py): session-pinned facts with dates preserved, re-stored across sessions.
- **Personal memory**: runtime/personal_memory_*, persona_updater.

Four SQLite files under artifacts/db/ (*runtime*): user.sqlite3 (memory, conversation, knowledge graph, learning tables), agent.sqlite3 (self-improvement records and agent metrics), system_index.sqlite3 (OS index) and coding_memory.sqlite3 (coding bug fixes). The blank user.sqlite3 template holds 27 tables plus three FTS5 indexes: memories, memories_archive, memory_meta, semantic, conversation_turns, conversations, session_summaries, kg_entities, kg_relations, recall_log, runtime_events, learning_replay, observations, habits, habit_events, habit_rules, user_patterns, user_model, eli_stances, belief_revisions, corrections, failures, error_tracking, improvements, capability_proposals, news_articles, news_reflections. Self-improvement records (improvements, failures, corrections, capability_proposals, error_tracking) are written to agent.sqlite3; the copies in the user database are empty. Recall is hybrid: keyword, FTS5, FAISS, knowledge graph and recent conversation, merged and reranked (§7).

12. Perception (eli/perception)

- **Vision**: vision.py (model-agnostic vision-language, Moondream and Qwen2.5-VL hot-swap; the multimodal encoder is forced to CPU to avoid a CUDA clip segfault), analyze_image.py, ambient_vision.py, screen_locator.py, gaze_engine.py, analyze_csv.py, analyze_pdfs.py.

- **STT**: `local_whisper_stt.py` (faster-whisper, offline when the Net switch is off), `audio_stt.py`, `voice_worker_streaming.py`. Default wake phrases: “computer”, “hey computer”, “eli”, “hey eli” (`wakeword.py`); a music-bleed filter and an echo gate sit in front of the transcript.
 - **TTS**: `tts_router.py` (Piper voices), `tts_xtts.py` (optional neural voices).
 - **OS**: `os_controller.py` (app, window, keyboard, mouse), `ui_tree.py`, `desktop_capabilities.py`.
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13. Persona (`eli/cognition/persona*`)

- `persona.txt` (authored) and `persona.auto.txt` (overlay) hold the voice.
 - `persona_updater.py` updates the overlay, knowledge graph and user profile each turn.
 - `persona.py`, `persona_values.py`, `persona_status.py`, `persona_hygiene.py`.
 - `tone_analyzer.py` and `tone_adaptor.py` read the user’s tone; `emotion_timeline.py` keeps a history that lets ELI notice a sustained mood.
 - Generation injects a budget-trimmed persona and situation brief (`engine.py::_build_enhanced_system`); a live-runtime fact is injected for model and identity questions so ELI reports the loaded model.
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14. Daemons and background loops

- **Proactive daemon** (`planning/proactive_daemon.py`): pattern signals (time habit, topic focus, recurring errors, active project); runs an autonomy tick every 30 minutes.
 - **Self-improvement loop** (`runtime/self_improvement.py`): started at boot, analyses failures immediately and then every 24 hours. Repair by the coding agent is opt-in and bounded.
 - **Habit scheduler** (`planning/habits_scheduler.py`, `habits.py`).
 - **Scheduler** (`kernel/scheduler.py`); **background tasks** (`runtime/background_tasks.py`).
 - **Code monitor** (`runtime/code_monitor.py`); **ambient vision loop**.
 - **World event bus** (`world/world_event_bus.py`).
 - **Memory upkeep**: once a day, started asynchronously at engine init and after responses.
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15. Learning and self-training (`eli/learning`)

A LoRA fine-tuning pipeline: `bootstrap_phi3_base.py`, `base_model_resolver.py`, `dataset_builder.py`, `dataset_filters.py`, `merge_reviewed_datasets.py`, `export_trainable_dataset.py`, `training_preflight.py`, `lora_trainer.py`, `lora_trainer_guard.py`, `lora_eval.py`, plus `target_registry.py` (operator-declared targets, any model family) and `review_queue.py` (the human review gate).

16. EliWorld (`eli/world`, the **EliWorld tab**, `kernel/world_model.py`)

An internal world the agent inhabits. `world_event_bus.py` receives runtime events; `local_world_bridge.py` bridges to the model; snapshots, ledger and journal are under `artifacts/world/`. Experimental.

17. Core infrastructure (`eli/core`)

- **netguard.py**: offline by default: `guarded_urlopen`, a process-wide socket guard (fail-closed on non-loopback while offline), and `allow_network()` for deliberate user-initiated fetches.
- **paths.py**, **portable_paths.py**, **legacy_paths.py**, **db_paths.py**: path resolution (dev tree vs platform directories, `ELI_*` overrides, read-only frozen bundles).

- **runtime_settings.py, config.py:** settings load, merge and heal; DEFAULTS are clean (offline, no model, wizard on). config/settings.json is per user and gitignored, seeded from config/templates/settings.template.json.
 - **hardware_profile.py, startupHardwareOptimizer.py, dynamic_runtime_budget.py:** detect GPU and VRAM and pick context, GPU layers and batch (one fit calculation shared by the loader and the startup dialog).
 - **model_download.py:** curated GGUF downloader (7 catalogue entries, resumable, magic and size validated, netguard-gated). Install-time menu only.
 - **grounding.py:** shared grounding helpers.
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18. GUI (eli/gui)

PySide6 (LGPL). eli_pro_audio_gui_v2_0.py (main window, 13.1k lines), app.py (boot), labs_tab.py (Labs, 5.7k lines), panels/ (startup.py: StartupModelSelectionDialog, FirstBootWizard and the hardware tuning dock; settings.py, agent_wizard.py, permission_dialog.py), tabs/ (eli_world_tab.py, experimental_tab.py, marketplace_tab.py, tasks_tab.py, training_tab.py), docks/ (operator console, proactive dock), widgets/ (animated face, voice downloader, Ollama selector). qt_compat.py allows a PyQt fallback from source.

19. On-disk layout (artifacts/, runtime)

```
artifacts/
├─ db/{user,agent,system_index,coding_memory}.sqlite3
├─ vectors/index.faiss           # semantic index
├─ conversations/               # conversation exports
├─ documents/ , scripts/        # generated artifacts
├─ runtime/                     # runtime state and capability inventory
├─ runtime_snapshot.json        # live model/runtime truth
├─ world/                       # snapshots, ledger, journal
├─ analyze_image_*/             # vision outputs
config/  settings.json (gitignored) · templates/ · plugins_state.json · voices/ ...
models/  <your>.gguf · embeddings/ · whisper/ · image/
tts_piper/ Piper voice files shipped with the repo
```

20. Known seams and weak points

1. **Large files.** engine.py, executor_enhanced.py, the main GUI file and the router hold about a third of the code. The executor is an action ladder with a high regression surface.
 2. **Internal state → spoken output.** The recurring failure class. The grounding gate and escalation defend it; they do not close it on the plain CHAT path.
 3. **Swallowed errors.** 160 handlers are a bare except: pass and 4,348 catch Exception, most of them logging at debug level. A ratchet test (tests/claims/test_no_silent_swallow.py) stops the silent count rising.
 4. **Latency versus model size.** On an 8 GB GPU a 24B model at Q5 offloads few layers and answers in minutes; a 7-9B Q4 is the sweet spot. The startup dialog shows the real fit.
 5. **Two GUI monoliths.** eli/gui/eli_pro_audio_gui_v2_0.py and eli/gui/labs_tab.py are single files of 13.1k and 5.7k lines.
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21. Where to make a change

To change...	Edit...
how input is routed to an action	eli/execution/router_enhanced.py
what an action does	eli/execution/executor_enhanced.py
which agents run and how grounded	eli/cognition/agent_bus.py
the 12-stage orchestration	eli/cognition/orchestrator.py
prompt and persona budget, the engine spine	eli/kernel/engine.py
anti-confabulation	eli/runtime/grounding_escalation.py, deterministic_grounding_gate.py
final answer shaping	eli/runtime/*response*, cognition/output_governor.py
memory store and recall	eli/memory/memory.py, policy.py, retrieval.py, vector_store.py
model loading and inference	eli/cognition/gguf_inference.py, inference_broker.py
offline and network policy	eli/core/netguard.py
paths, settings, model download	eli/core/{paths, runtime_settings, model_download}.py
the GUI	eli/gui/eli_pro_audio_gui_v2_0.py, eli/gui/panels/, eli/gui/tabs/
evaluation and regression	tools/eval/

Plugins, agent specifications and training surfaces

eli/plugins/: a loader plus a gated marketplace client

Module	Lines	Role
manager.py	708	discovery, enable and disable, install and uninstall
permissions.py	395	capability vocabulary, consent decisions, grants, audit ledger
manifest.py	302	manifest schema and static capability verification against source
integrity.py	291	sha256 pinning, ed25519 signatures, operator-trusted publishers
security_scan.py	547	multi-engine malware scanner
marketplace.py	793	federated registries, preview and install, licence keys
mcp.py	481	MCP server config, runtime preflight, handshake verification

`_plugins_dir()` resolves through `paths.plugins_dir()`, and `_plugin_search_dirs()` scans bundled built-ins and user installs, so a downloaded plugin is never written into a read-only installation.

eli/cognition/: agents get a specification

Module	Lines	Role
agent_spec.py	436	AgentSpec: objective, prompt, triggers, success criteria, examples
agent_trust.py	262	path-keyed trust with provenance, scanning, revocation

agent_bus.SpecAgent runs a spec against the local model and scores the output against the spec's own criteria. Spec agents execute no arbitrary code and bypass the trust chain.

GUI surfaces

tabs/training_tab.py (Labs ► Training), tabs/marketplace_tab.py (Settings ► Marketplace), panels/permission_dialog.py (consent dialog and worker-to-GUI-thread bridge).

Paths

paths.learning_dir() joins models_dir(), voices_dir() and plugins_dir(). The rule: runtime state never lives in the installation, because project_root() is a read-only mount in a packaged build.